

Market efficiency and the welfare theorems



Adam Smith claimed that a system of free markets spontaneously leads to a socially desirable allocation of goods (see [The invisible hand](#)). Without any central organization, the right goods and services get produced and end up in the hands of those who can put them to best use. It's as if all the buyers and sellers, workers and employers are guided by an 'invisible hand'. In the 20th century, economists investigated this idea mathematically, deriving 'welfare theorems' that set out conditions under which free markets lead to social good.

Their criteria for social good was Pareto efficiency (see [Pareto efficiency](#)): an allocation is said to be Pareto efficient if no further reallocations of goods can be made to benefit at least one person while hurting no one. One of the welfare theorems states

that under certain conditions any allocation of goods arrived at through the action of free markets is Pareto efficient. The catch is that the conditions are so stringent that they are unlikely to hold in the real world.